NATURAL LANGUAGE PROCESSING

Course	B.Tech.-VI-Sem.	L	T	P	C
Subject Code	22CAPC63	3	-	-	3

Course Outcomes (COs) & CO-PO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	Upon completion of course the students will be able to	PO2	PO3	PO8	PO12	PSO1
CO1	explain fundamentals of NLP and morphology	3	2	3	3	3
CO2	demonstrate word level statements and syntactic analysis	3	2	3	3	3
CO3	make use of context free grammar and parsing techniques	3	3	3	3	3
CO4	apply semantic analysis techniques to solve various problems	3	3	3	3	3
CO5	illustrate language generation and discourse analysis	3	2	3	3	3

Syllabus

Unit	Title/Topics	Hours
I	**Overview and Morphology**	9
Introduction: Models and Algorithms - Regular Expressions - Basic Regular Expression Patterns - Finite State Automata. **Morphology:** Inflectional Morphology - Derivational Morphology - Finite-State Morphological Parsing -Porter Stemmer.		
II	**Word Level and Syntactic Analysis**	10
N-grams Models of Syntax - Counting Words - Unsmoothed N- grams, Smoothing- Backoff Deleted Interpolation – Entropy - English Word Classes - Tagsets for English, Part of Speech Tagging-Rule Based Part of Speech Tagging - Stochastic Part of Speech Tagging - Transformation-Based Tagging.		
III	**Context Free Grammars and Parsing**	5+4=9
PART-A: Context Free Grammars for English Syntax- Context- Free Rules and Trees – Sentence-Level Constructions– Agreement – Sub Categorization.		
PART-B: Parsing – Top-down – Earley Parsing - feature Structures – Probabilistic Context-Free Grammars.		
IV	**Semantic Analysis**	10
Representing Meaning - Meaning Structure of Language - First Order Predicate Calculus; Representing Linguistically Relevant Concepts -Syntax- Driven Semantic Analysis - Semantic Attachments -Syntax- Driven Analyzer; Robust Analysis - Lexemes and Their Senses - Internal Structure - Word Sense Disambiguation -Information Retrieval		
V	**Language Generation and Discourse Analysis**	10
Discourse -Reference Resolution - Text Coherence - Discourse Structure - Coherence; Dialog and Conversational Agents - Dialog Acts - Interpretation - Conversational Agents - Language Generation – Architecture - Surface Realizations - Discourse Planning; Machine Translation - Transfer Metaphor–Interlingua – Statistical Approaches.		
Textbooks		
1. Speech and Language Processing, Daniel Jurafsky and James H. Martin, , Prentice Hall; 2nd Edition, 2008. 2. Foundations of Statistical Natural Language Processing, Christopher D. Manning and Hinrich Schuetze, MIT Press, 1999.		
References		
1. James Allen, Natural Language Understanding, Addison Wesley; 2nd Edition, 1994.		